

## TITLE PAGE

- **Problem Statement ID-** 26065
- **Problem Statement Title-** Autonomous Low-Cost Ocean Observation Platform for Polar and Southern Oceans
- **Theme-** Robotics and Drones
- **PS Category-** Hardware
- **Team ID-**
- **Team Name-** FusionX

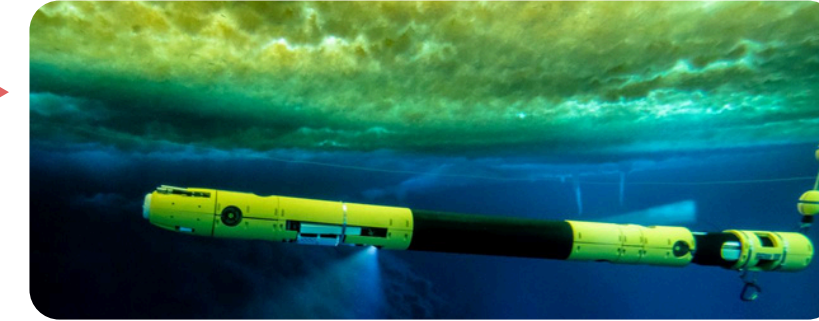


## Problem

- **AQUILA: An Adaptive Autonomous Ocean Observation and Intelligence Platform with AI-Enabled Underwater Anomaly Detection.**



Under-ice vehicle



## Addresses

- **Ice-Blocked Surfacing:** Decides when not to surface
- **Under Ice-Gaps:** Stores observations until contact returns
- **Fixed-Sampling:** Increases sampling only during detected change
- **Underwater Anomalies:** Detects marine debris -> Side-Scan Sonar + AI



Sea-ice buoy

## Solution

- **Predict the Ice:** Detects rising ice risk and aborts unsafe surfacing.
- **Detect Change:** Detects ocean shifts → increases sampling.
- **Lose the Link:** Stores data underwater → transmits when connected.
- **Sonar-to-Insight:** Converts Side-Scan Sonar imagery into AI-based marine-debris detections.

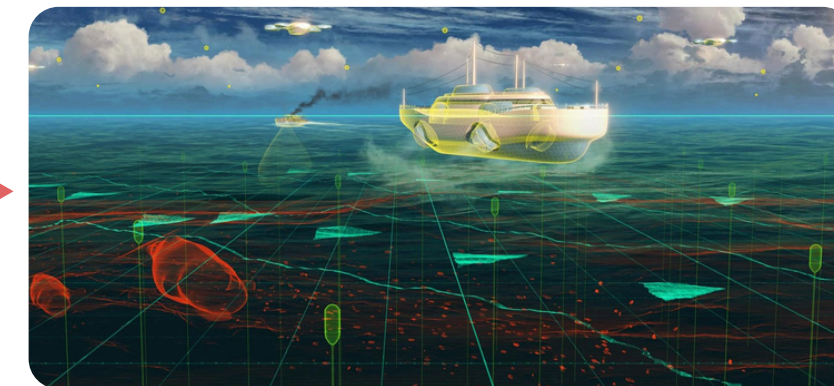


Autonomous Ocean Profiling



## Innovation

- **Our Innovation is a Digital-Twin-driven Edge-AI mission engine enabling self-adaptive, ice-aware ocean observation.**



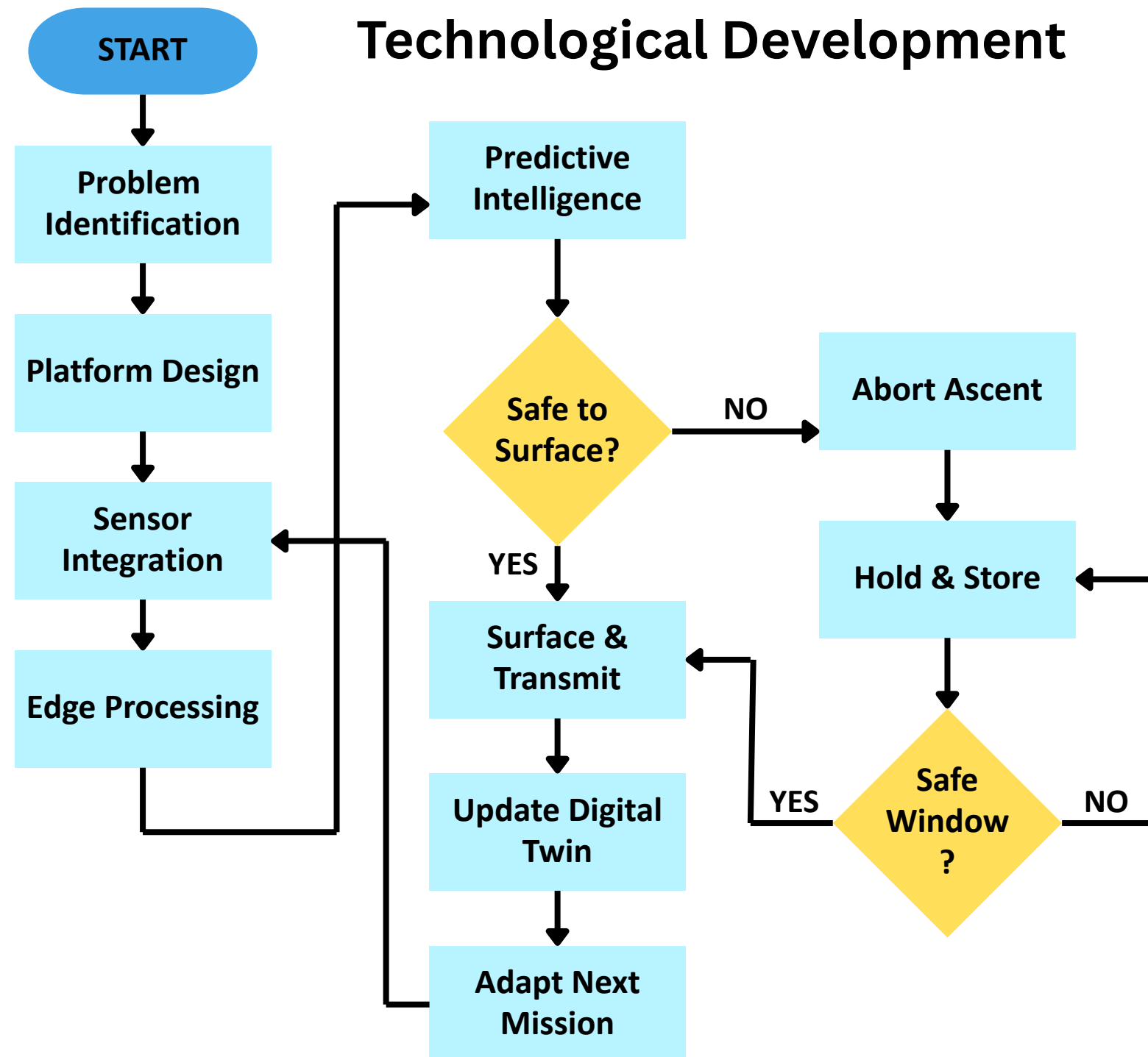
Digital Twin + Edge AI

 Digital Twin →  Edge AI →  Adaptive Mission



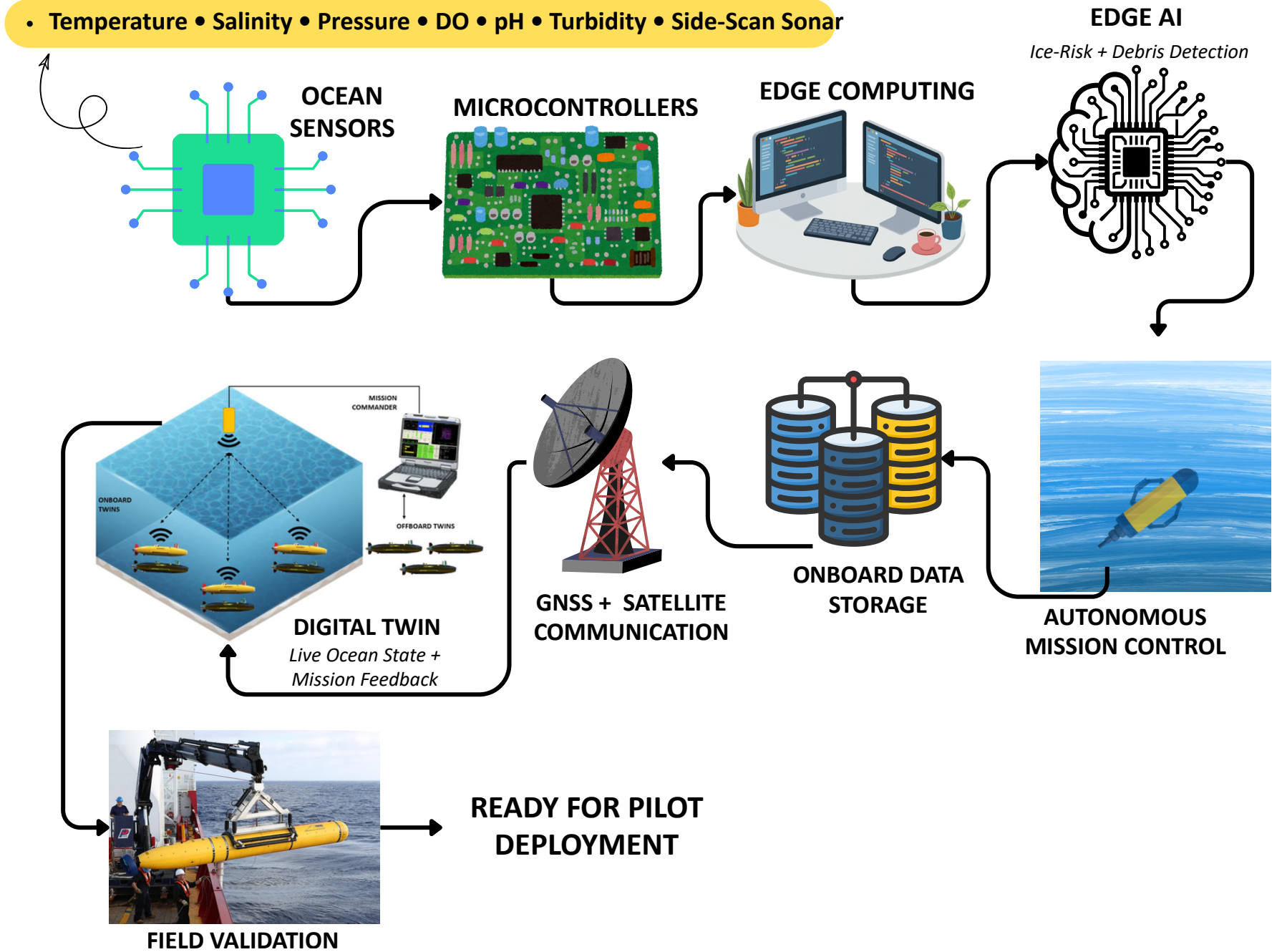
## TECHNOLOGIES

We are using smart sensing, Side-Scan Sonar, Edge AI, autonomous control, satellite communication, and Digital Twin technology.



## METHODOLOGY

We follow a sense–process–predict–detect–decide–adapt approach for autonomous ocean observation.

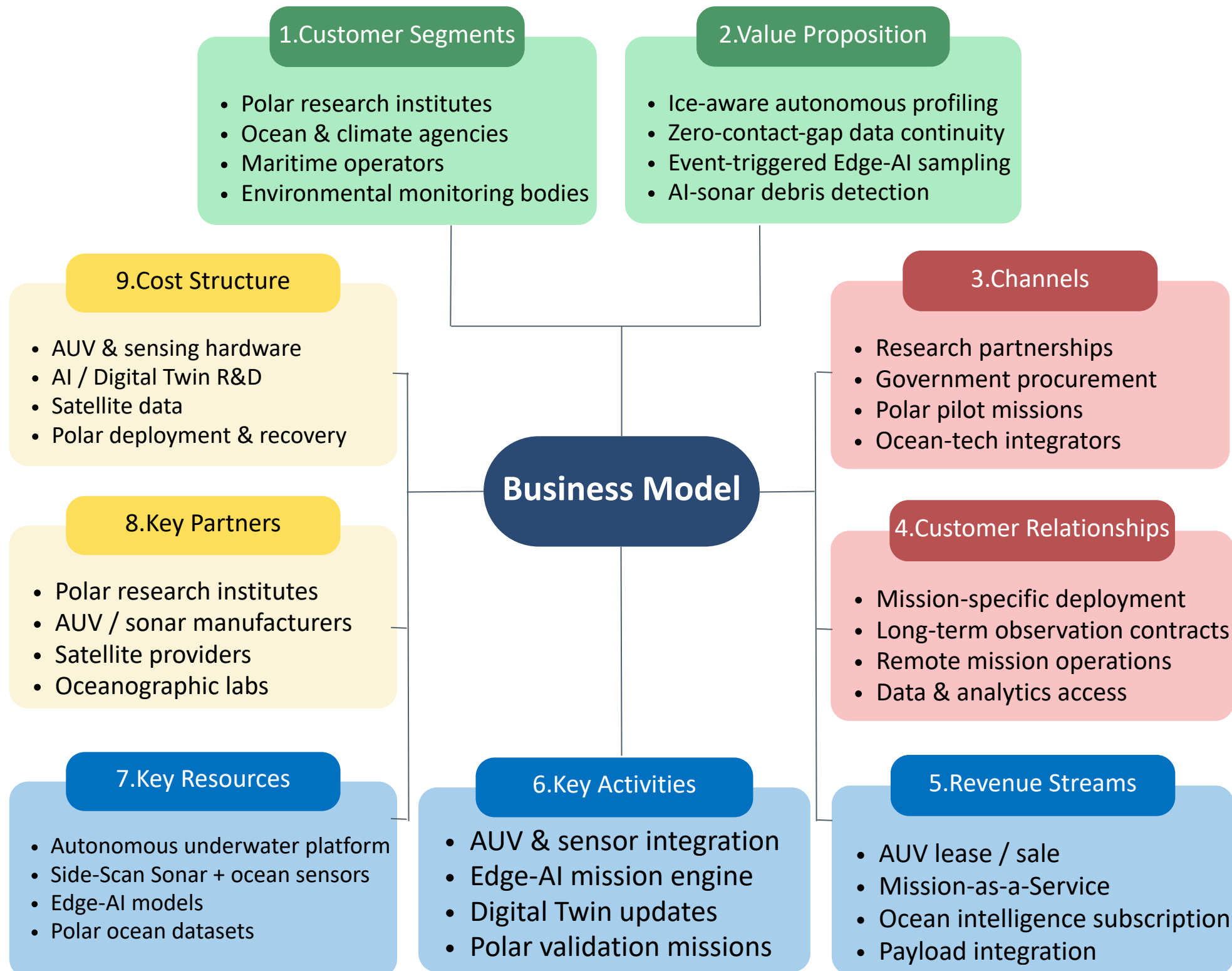


## Challenges:

Harsh polar conditions, intermittent connectivity, and limited underwater visibility challenge reliable long-duration autonomous observation.

## Strategies:

We focused on sensing smarter, deciding locally, adapting to changing ocean conditions, and staying operational through ice and communication gaps.



## STRENGTHS

- Ice-aware autonomy** — safer surfacing decisions
- Communication resilience** — stores data during blackouts
- Adaptive Edge-AI** — responds to ocean changes
- Sonar + AI fusion** — detects underwater debris

## WEAKNESS

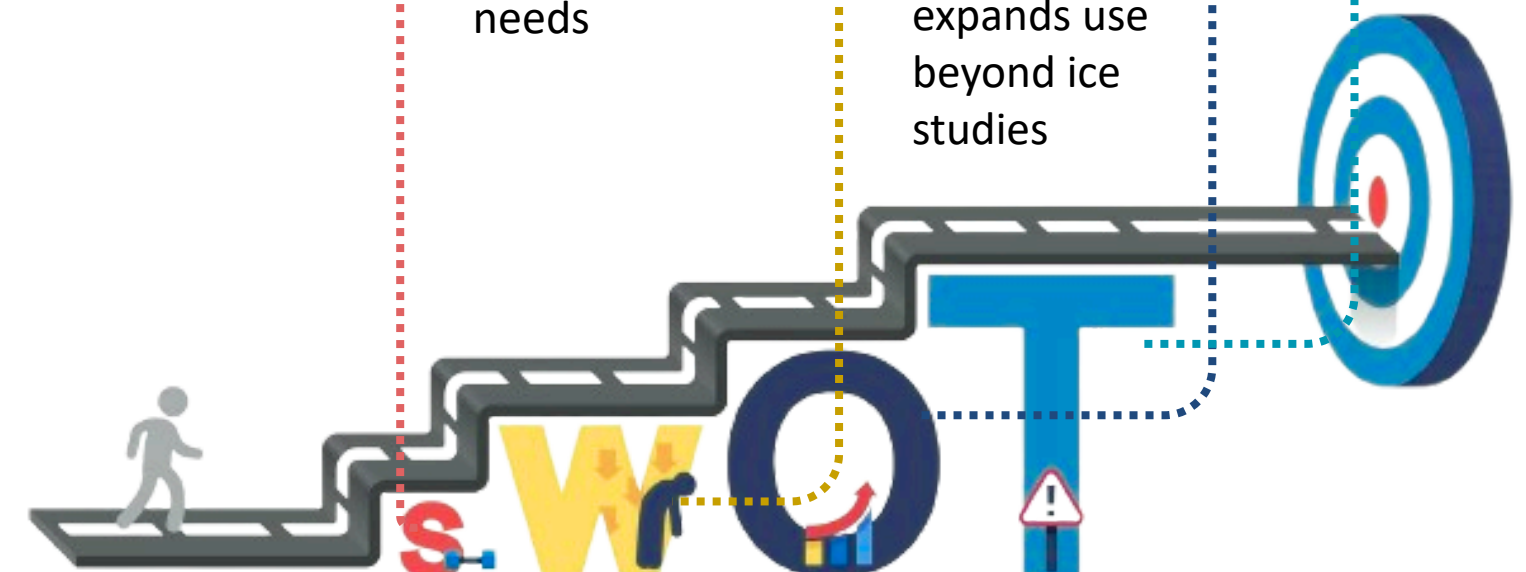
- Power-constrained computing** — limits onboard AI complexity
- Polar recovery logistics** — specialized deployment required
- Harsh-environment hardware** — higher maintenance needs

## OPPORTUNITIES

- Persistent polar monitoring** — fewer human interventions
- Ocean intelligence** — valuable long-term datasets
- Fleet-scale deployment** — wider ocean coverage
- Marine-debris monitoring** — expands use beyond ice studies

## THREATS

- Extreme ice conditions** — unpredictable operating environment
- Communication blackouts** — delayed data transmission
- Hardware degradation** — prolonged underwater exposure
- Competing AUV platforms** — established alternatives





# IMPACT AND BENEFITS

## Impacts and Benefits

This innovative solution transforms **challenging ocean environments** into opportunities for **safer exploration, continuous data, & scalable ocean intelligence**, opening new possibilities across **polar, climate, & marine monitoring**.



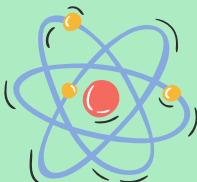
### ECONOMIC:

- Fewer ship-dependent surveys
- Lower human intervention
- Longer autonomous missions



### ENVIRONMENTAL:

- Continuous ice & ocean monitoring
- AI-powered marine-debris detection
- Adaptive sampling reduces wasted energy



### SCIENTIFIC:

- Persistent polar datasets
- Real-time mission adaptation
- Digital Twin-ready ocean intelligence

**PROTOTYPE COST BREAKDOWN:** Waterproof Hull/Enclosure — ₹2,000 | ESP32/STM32 — ₹1,000 | IMU — ₹500 | Depth Sensor — ₹1,500 | Temperature Sensor — ₹300 | Salinity Sensor — ₹1,500 | Buoyancy Actuator — ₹1,500 | Propulsion — ₹1,500 | Battery + BMS — ₹1,500 | GPS/GNSS — ₹500 | LoRa/Telemetry — ₹500 | Waterproof Connectors — ₹1,000 | Frame/Ballast — ₹1,500 | SSS Demonstration — ₹2,000 | Edge Processing — ₹1,500 | Testing/Misc. — ₹1,700 **TOTAL EST. ≈ ₹19.5K**

## SCALABILITY: FROM PROTOTYPE TO DEPLOYMENT

### Prototype

- ≤ ₹20K — Working demonstrator
- Depth + Temp. + Salinity — Core sensing
- SSS + AI — Anomaly detection demo
- Sense → Decide → Adapt — Autonomous loop
- Tank / Shallow Water — Initial validation

### MVP

- ₹8–14L — Field-ready prototype
- Scientific sensors — Improved ocean measurements
- SSS + Edge AI — Integrated anomaly detection
- 500–1,000 m — Target depth
- Months — Extended deployment

### Final Product

- ₹23–33L — Polar deployment target
- Scientific-grade sensing — Persistent observation
- Marine-grade SSS + Onboard AI — Autonomous investigation
- ~2,000 m — Target depth
- 6+ months — Long-duration operation

| What Matters     | Existing approach         | Our solution- AQUILA                |
|------------------|---------------------------|-------------------------------------|
| Deployment       | Ship-dependent surveys    | Launch once → observe autonomously  |
| Monitoring       | Periodic data collection  | Persistent ocean observation        |
| Anomaly Response | Find it after the mission | Detect → investigate while deployed |
| SSS              | No on-demand SSS          | SSS wakes only when needed          |
| Intelligence     | Analysis after recovery   | AI scores & classifies onboard      |
| Connectivity     | Data-link dependent       | Store → Resend when link returns    |

# RESEARCH AND REFERENCES



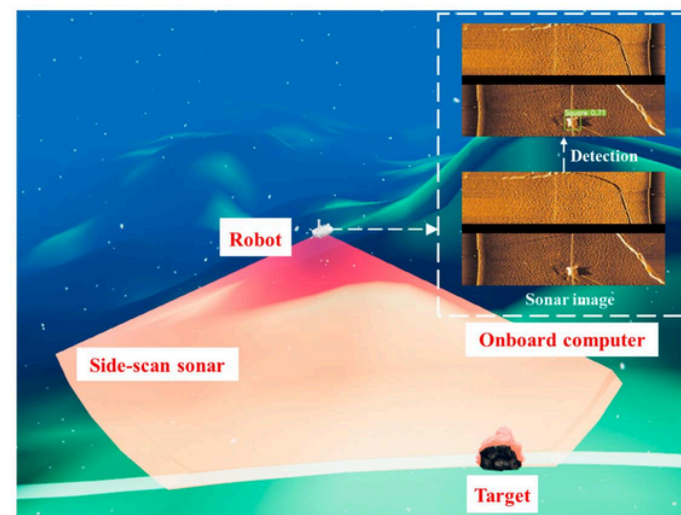
## RESEARCH:

Our research focused on making underwater observation **adaptive under ice, resilient to communication loss**, and capable of **detecting anomalies without waiting for post-mission analysis**. We studied polar AUV path planning, side-scan sonar anomaly detection, and Digital Twin-based AUV control to shape **AQUILA's sense–predict–decide–adapt architecture**.

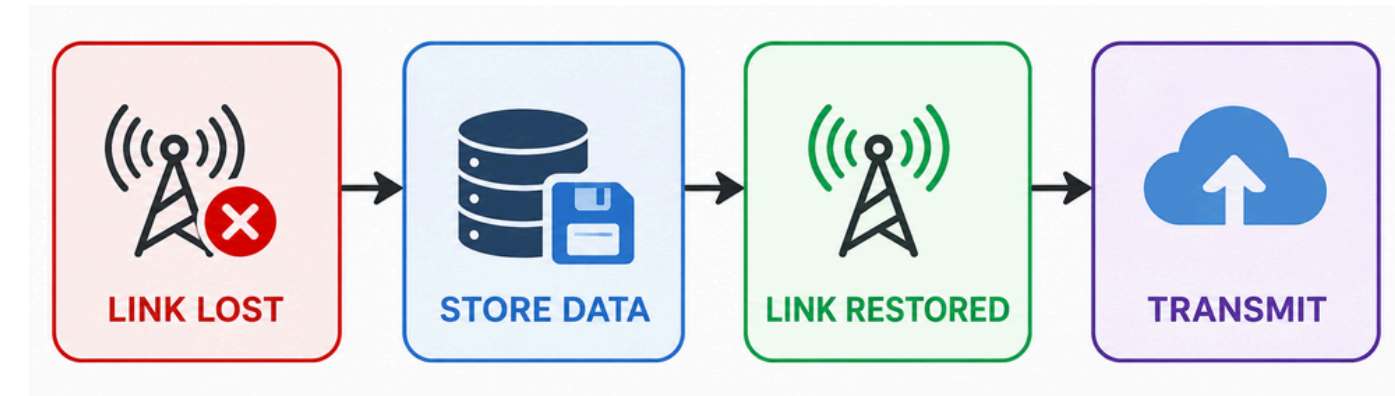
## SYSTEM VALIDATION



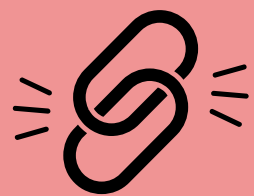
Under-Ice Autonomous Observation



Side-Scan Sonar → AI Anomaly Detection



Communication-Resilient Data Handling



## REFERENCES:

1. Partridge et al. (2026) — A Digital Twin Ocean: Can We Improve Coastal Ocean Forecasts Using Targeted Marine Autonomy? — Ocean Science.
2. Coffelt, J. P. & Christensen, J. H. (2021) — Anomaly Detection in Side-Scan Sonar. — OCEANS 2021, IEEE.
3. Yan et al. (2026) — Digital Twin-Driven Swarm of Autonomous Underwater Vehicles for Marine Exploration. — Communications Engineering.